

VORU, Estonia

WMO: 262490

Lat: 57.8464N

Lon: 27.0194E

Elev: 84

StdP: 100.33

Time Zone: 2.00 (EUE)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.9	-19.0	-25.5	0.4	-22.9	-21.5	0.6	-18.5	7.7	2.8	6.9	1.7	0.9	80	0.540

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.1	28.4	19.3	26.6	18.3	24.9	17.6	20.6	26.0	19.6	24.5	18.6	23.2	2.9	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.7	13.7	22.8	17.7	12.8	21.9	16.7	12.0	20.9	59.5	26.1	56.1	24.6	52.9	23.2	24.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.4	5.8	5.2	-25.8	30.9	5.2	1.7	-29.6	32.1	-32.6	33.1	-35.5	34.0	-39.3	35.3		
(5)			-25.8	21.9	4.8	1.0	-29.3	22.7	-32.1	23.3	-34.8	23.9	-38.3	24.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	6.5	-5.3	-4.8	-0.5	5.7	12.1	15.6	18.6	17.1	12.0	5.9	1.8	-1.4	(6)
(7)		DBStd	9.39	7.00	6.76	4.96	3.86	4.23	3.34	3.16	3.08	3.27	4.18	4.16	5.27	(7)
(8)		HDD10.0	2136	476	414	327	138	24	2	0	0	17	137	247	353	(8)
(9)		HDD18.3	4420	734	648	586	380	197	96	36	61	189	386	497	612	(9)
(10)		CDD10.0	841	0	0	0	8	91	168	266	221	78	10	0	0	(10)
(11)		CDD18.3	83	0	0	0	0	5	12	43	23	1	0	0	0	(11)
(12)		CDH23.3	733	0	0	0	1	59	121	357	191	3	0	0	0	(12)
(13)		CDH26.7	150	0	0	0	0	8	19	83	39	0	0	0	0	(13)
(14)	Wind	WSAvg	2.5	2.8	2.6	2.7	2.6	2.4	2.4	2.1	2.1	2.2	2.5	2.7	3.0	(14)
(15)	Precipitation	PrecAvg	611	33	28	27	35	50	72	78	82	63	57	52	45	(15)
(16)		PrecMax	832	82	64	69	109	114	151	194	213	155	120	104	89	(16)
(17)		PrecMin	447	5	4	8	2	9	11	14	11	12	13	14	13	(17)
(18)		PrecStd	97	16	14	15	26	26	41	45	51	31	29	21	18	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.0	6.1	14.2	21.7	27.8	29.0	31.0	30.0	23.6	16.7	10.9	9.2	(19)
(20)			MCWB	6.0	4.4	6.9	11.3	18.1	19.1	20.6	20.3	16.5	13.3	8.4	8.2	(20)
(21)		2%	DB	5.0	4.7	10.4	18.3	24.9	26.5	29.0	27.5	21.1	14.1	8.8	7.3	(21)
(22)			MCWB	4.0	3.4	5.4	9.9	16.3	17.2	19.8	19.0	15.5	12.0	7.5	6.1	(22)
(23)		5%	DB	3.4	3.5	7.5	15.9	22.5	24.5	27.1	25.4	19.1	12.9	8.0	5.7	(23)
(24)			MCWB	2.5	2.4	4.0	8.5	14.5	16.5	19.2	18.3	14.6	11.1	7.0	4.7	(24)
(25)		10%	DB	2.2	2.5	5.5	13.3	20.2	22.2	25.2	23.4	17.4	11.8	7.0	4.3	(25)
(26)			MCWB	1.5	1.6	3.1	7.2	13.1	15.5	18.3	17.3	13.8	10.3	6.0	3.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.1	4.8	7.9	12.3	19.5	20.5	22.0	21.8	18.1	13.7	9.0	8.2	(27)
(28)			MCDWB	6.9	5.7	13.0	18.9	25.5	25.7	28.2	27.5	21.6	15.5	10.1	9.1	(28)
(29)		2%	WB	4.1	3.5	5.7	10.5	17.3	18.7	20.9	20.2	16.4	12.4	7.9	6.2	(29)
(30)			MCDWB	4.8	4.4	9.0	16.6	22.7	24.1	26.5	24.9	19.4	13.9	8.6	7.0	(30)
(31)		5%	WB	2.7	2.5	4.5	9.3	15.7	17.4	20.0	19.0	15.2	11.6	7.0	4.8	(31)
(32)			MCDWB	3.2	3.3	6.8	14.5	20.6	22.4	25.4	23.4	18.1	12.8	7.8	5.5	(32)
(33)		10%	WB	1.6	1.6	3.4	8.0	14.3	16.3	19.1	18.0	14.2	10.5	6.1	3.6	(33)
(34)			MCDWB	2.1	2.4	5.3	12.7	19.0	20.9	23.9	22.5	16.7	11.6	6.8	4.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.6	5.5	7.3	9.8	10.6	9.2	9.1	7.6	5.5	3.5	3.5	(35)	
(36)			MCDBR	3.4	4.2	9.8	14.8	14.0	12.4	12.3	12.7	10.5	6.4	4.4	3.7	(36)
(37)			MCWBR	3.4	3.4	6.1	7.6	6.4	5.0	4.7	5.4	5.7	4.6	3.9	3.4	(37)
(38)		5% WB	MCDWR	3.4	3.9	8.3	12.3	11.8	10.1	11.0	10.3	8.9	5.9	4.2	3.4	(38)
(39)			MCWBR	3.4	3.5	5.7	7.0	6.1	5.1	4.7	5.1	5.3	4.6	3.9	3.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.275	0.307	0.337	0.377	0.371	0.369	0.396	0.393	0.362	0.337	0.316	0.270	(40)	
(41)		taud	2.122	2.124	2.170	2.278	2.368	2.405	2.348	2.352	2.418	2.394	2.221	2.120	(41)	
(42)		Ebn at Noon	556	709	799	830	863	869	833	807	778	688	505	443	(42)	
(43)		Edh at Noon	53	84	106	113	110	108	112	104	84	65	49	39	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.41	1.11	2.53	3.92	5.26	5.42	5.20	4.14	2.65	1.18	0.39	0.25	(44)	
(45)		RadStd	0.08	0.20	0.32	0.47	0.56	0.45	0.53	0.42	0.31	0.18	0.06	0.04	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (28 neighbors)	+0.42	N/A	N/A	N/A	N/A	N/A	-166	-164	N/A	N/A

Nomenclature: See separate page